Date No chapter 2

صلي على النبي
بسم الله

① Boyle's Law at constant $\boxed{\text{Tem}}$

$V \propto \frac{1}{P}$ $P_1V_1 = P_2V_2$

② Avogadros Law molar volums are same at constant $\boxed{\text{press}}$ and $\boxed{\text{Tem}}$

$V = \frac{V}{n}$ $\frac{V_1}{n_1} = \frac{V_2}{n_2}$ Charles Law at con press $\boxed{V \propto T}$

③ $PV = nRT \rightarrow$ idel

$\hookrightarrow \Rightarrow 0.082$ at $V \rightarrow L$, $P \rightarrow atm$

④ mole fraction and partial perssure

$X_i = \frac{P_i}{P_{total}}$ $P_i = X_i \, P_{total}$

$\hookrightarrow$ $*\, 1 = X_i$ مجموع كل

مجموعة مركبات يعني

⑤ $K_e = \frac{3}{2} RT$

$J \hookleftarrow$ $\hookrightarrow 8.314$

⑥ $\bar{u} = \sqrt{\frac{3RT}{M}}$ $\frac{\bar{u}_b}{\bar{u}_a} = \sqrt{\frac{m_a}{m_b}}$

Kg $\leftarrow M \rightarrow$ molecular mass

⑦ $\left(P_{obs} + \frac{an^2}{V^2}\right) + (V - nb) = nRT \rightarrow$ Real

$P_{obs} = \left(P - \frac{an^2}{V^2}\right)$ C.F

Chapter 1

$$T_K = T_C + 273.15$$

$$T_C = (T_F - 32°F) \times \frac{5C°}{9°F}$$

$$T_F = \left(T_C \times \frac{9°F}{5C°}\right) + 32°F$$

conversion factor = $\frac{\text{unit required}}{\text{unit given}}$

Chapter 3

$$\text{viscosity } \eta = \frac{F \times d}{V \times A} = \boxed{\text{Poise}}$$

$$\eta = \frac{\pi r^4 P t}{8 l V}$$

$$\frac{\eta_1}{\eta_2} = \frac{t_1 d_1}{t_2 d_2}$$

$$\eta_{specific} = \frac{\eta_{liq}}{\eta_{H_2O}}$$

$$P.M = dRT$$

$$P.V = \frac{m}{M} RT$$

* boiling point
↑ in London force (non polar)

bP ∝ Polarizability

∝ ↑ size

The stronger ←

F_2
Cl_2
Br_2

Chapter 4

$$\ln\left(\frac{P_1}{P_2}\right) = \frac{-\Delta H_v}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Note Normal Boiling Point

1 atm = 760 torr = 760 mmHg

Factors affecting on P_v

1. Molecular mass → MM ↑ P_v ↓
2. Intermolecular force → f ↑ P_v ↓
3. Temperature → T ↑ P_v ↑

Highest volatile = Lowest Boiling Point

Chapter 5

$\Delta E = q + w$ — $\Delta E = \Delta H + w$

I.E — Heat — Work

ΔE
- Absorbed → +ve Into
- Emitted → −ve out

q
- Exothermic → −ve Lose
- Endothermic → +ve Gain

w
- on system or By surround → +ve
- By system or on surround → −ve

Chapter 6

Rault's Law:

Solid X Liquid

$$P_{soln} = X_{solvent} \cdot P^{\circ}_{solvent}$$

Liquid X Liquid

$$P_{soln} = (P^{0}_{A} \cdot X_{A}) + (P^{0}_{B} \cdot X_{B})$$

$$i = \frac{\text{Moles of Particles in Soln}}{\text{Moles of dissolved solute}}$$

EX $NaCl \longrightarrow Na^{+} + Cl^{-}$

$i = 2$

$i = 1 \Longleftarrow$ Glucose زي لو حاجه مش بتتفكك *

Boiling Point:

$$T_{b(solvent)} < T_{b(solution)}$$

$$\Delta T_b = \underbrace{i}_{\text{vant Hoff}} \underbrace{K_b}_{\text{Constant No.}} \overbrace{m_{solute}}^{\text{molality}} = T_{b(soln)} - T_{b(solvent)}$$

* Aqueous Soln. ⇝ Water $T_b = 100 C^{\circ}$

$T_f = 0 C^{\circ}$

Freezing Point:

$$T_{f\ solution} < T_{(f)\ solvent}$$

$$\Delta T_f = -i K_f m_{solute} = T_{f\ soln.} - T_{f\ solvent}$$

OR

$$\Delta T_f = i K_f m_{solute} = T_{f\ solvent} - T_{f\ soln.}$$

osmotic Pressure :

$$\pi = MRTi = \frac{n}{V}RTi = \frac{m}{M_w \cdot V} RTi$$

$M \rightarrow$ molarity $R \rightarrow 0.082$ $T \rightarrow$ Kelvin

concetration :

1. Mole fraction (X_i)

$$X_i = \frac{n_i}{n_t}$$

$t \rightarrow$ Total

2. mass fraction

$$MF = \frac{W_i}{W_t}$$

3. Molarity (M)

$$M = \frac{n_{solute}}{V_{solution}}$$ mol/L, Molar, M

4. Molality (m)

$$m = \frac{n_{solute}}{kg \rightarrow W_{solvent}}$$ mol/kg, molal, m

5. Normality (N)

$$N = \frac{n_{(eq)}\ solute}{V_{solution}}$$

$n_{(eq)} = n \cdot X$ — no. of ele, H^+, OH^-

$$\text{equivelant weight} = \frac{M_w}{X}$$

Chapter 7

$$K_c = \frac{[P]^n}{[R]^m}$$

$$K_p = \frac{[P]^n}{[R]^m}$$

$P \rightarrow$ Product

$R \rightarrow$ Reactant

$n \rightarrow$ no. of mole of (P)

$m \rightarrow$ no. of mole of (R)

* The Solid and Liquid No effect on K_c OR K_p concentration

K_c OR K_p → No change except by change Temperature

Relation between (K_c) and (K_p)

$$K_p = K_c (RT)^{\Delta n_{gas}}$$

$\Delta n \rightarrow P - R$

$$K_c = K_p (RT)^{-\Delta n_{gas}}$$

$R + \text{Heat} \rightleftharpoons P$ Endothermic

$R \rightleftharpoons P + \text{Heat}$ Exothermic

$$2NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2NOCl_{(g)}$$

3 → 2

$V \rightarrow P$

$V \propto \frac{1}{P}$

↑ P increasing

↓ V decreasing

شوفه قالك ايه وامش مع السلام

Date No

Chapter 8

concentration of acid

$[H^+]^2 = K_a C_a$; $[H^+] = \sqrt{K_a C_a}$

$[OH^-]^2 = K_b C_b$; $[OH^-] = \sqrt{K_a C_a}$

$PH = -\log[H]$; $POH = -\log[OH]$

$PH + POH = 14$

$[H^+][OH^-] = 10^{-14} \rightarrow K_w$

Ionization oR dissociation $= \frac{[H^+]}{[C_a]}$ oR $\frac{[OH^-]}{[C_b]}$

* The week acid have stronge conjugate Base - والعكس

* $K_w = K_a K_b$

in buffere

$[H^+] = K_a \frac{[Salt]}{[acid]}$

$PH = -\log K_a + \log \frac{[Salt] \text{ oR } [con.base]}{[acid]}$

$\rightarrow PK_a$

$[OH^-] = K_b \frac{[Salt]}{[Base]}$

$POH = PK_b + \log \frac{[Salt] \text{ oR } [con\ acid]}{[Base]}$